

Quantum Entropies



- These properties enable quantum generalizations of classical quantities like conditional entropy, joint entropy, and mutual information.

**Joint
Entropy**

**Conditional
Entropy**

**Coherent
Information**

**Mutual
Information**



Joint Quantum Entropy

- **Definition:** Let $\rho_{AB} \in \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B)$ be the density operator of a bipartite system. The **joint entropy** is defined as:

$$S(AB)_\rho = -\text{Tr}(\rho_{AB} \log \rho_{AB})$$

- This is a natural extension of the quantum entropy to composite systems.
- For a multipartite state ρ_{ABC} , we define $S(AB)_{\rho_{ABC}} = S(\text{Tr}_C(\rho_{ABC}))$.
- **Properties:**
 1. Additivity of quantum entropy under tensor product: $S(A, B) = S(A) + S(B)$ if $\rho_{AB} = \rho_A \otimes \rho_B$
 2. Sub-additivity inequality: $S(A, B) \leq S(A) + S(B)$
 3. Triangle inequality: $S(A, B) \geq |S(A) - S(B)|$



Quantum vs Classical Joint Entropy

- **Classical case:** $H(X, Y) \geq H(X)$, $H(X, Y) \geq H(Y)$
- **Quantum case:** Not always!
 - Consider a pure bipartite state: $|\psi\rangle_{AB} = \sum_i \lambda_i |e_i\rangle_A |f_i\rangle_B$
 - Marginal states: $\rho_A = \sum_i \lambda_i^2 |e_i\rangle_A \langle e_i|$, $\rho_B = \sum_i \lambda_i^2 |f_i\rangle_B \langle f_i|$
 - Then, $S(AB)_\psi = 0$, $S(A)_\psi = S(\rho_A) \neq 0$, $S(B)_\psi = S(\rho_B) \neq 0$



Marginals of Pure States

- **Theorem:** Let $|\psi\rangle_{AB}$ be a pure bipartite state. Then:

$$S(A)_\psi = S(B)_\psi, \quad S(AB)_\psi = 0$$

- **Additivity for Product States:** $S(\rho_A \otimes \sigma_B) = S(\rho_A) + S(\sigma_B)$

- **Example:** Consider hybrid classical-quantum state: $\rho_{AZ} = \sum_z p_Z(z) \rho_z^A \otimes |z\rangle_Z \langle z|$

- Then: $S(AZ)_\rho = S(Z) + \sum_z p_Z(z) S(\rho_z^A)$

- **Example:** Bipartite Entangled State, Bell state: $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|0_A 0_B\rangle + |1_A 1_B\rangle)$.

- Marginal states: $\rho_A = \frac{1}{2}I, \quad \rho_B = \frac{1}{2}I$

- Entropies: $S(AB)_{\Phi^+} = 0, \quad S(A)_{\Phi^+} = S(B)_{\Phi^+} = \log 2 = 1$



Conditional Entropy

- **Definition:** The conditional entropy of a bipartite state ρ_{AB} is:

$$S(A | B) = S(AB) - S(B)$$

- **Classically:**

- Entropies are always non-negative
- Hence $S(A | B) \geq 0$
- No need to define coherent information

- **Quantumly:**

- $S(A | B)$ can be negative!
- Motivates the need for **coherent information**.

- **Example:** Bell State $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|0_A 0_B\rangle + |1_A 1_B\rangle)$.

- **Pure state** $|\Phi^+\rangle \Rightarrow S(AB) = 0$ and $S(A) = S(B) = 1$
- **Conditional Entropy:** $S(A | B) = S(AB) - S(B) = 0 - 1 = -1$



Entanglement and Conditional Entropy

- **Theorem:** A pure bipartite quantum state $|\psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ is entangled if and only if:

$$S(A|B)_\psi < 0$$

- Since $|\psi\rangle_{AB}$ is pure: $S(AB)_\psi = 0 \Rightarrow S(A|B)_\psi = -S(B)_\psi$
- So equivalent statement: $|\psi\rangle_{AB}$ is entangled $\iff S(B)_\psi > 0$
- **Proof:** $\Rightarrow |\psi\rangle_{AB}$ entangled $\Rightarrow$ Schmidt rank $d > 1$

- Bob's marginal state: $\rho_B = \sum_{i=1}^d \lambda_i^2 |i\rangle_B \langle i|$

- Then: ρ_B not pure and $S(\rho_B) > 0$

- **Proof:** $\Leftarrow$ Assume $S(B)_\psi > 0$, then $S(\rho_B) = -\sum_{i=1}^d \lambda_i^2 \log(\lambda_i^2)$

- If $d = 1$, then $\lambda^2 = 1 \Rightarrow S(B) = 0$

- So $S(B)_\psi > 0 \Rightarrow d > 1 \Rightarrow |\psi\rangle_{AB}$ is entangled



Why Coherent Information?

- **Issue:** Classical conditional entropy $H(A | B)$ can't be negative, but in quantum information theory it can.
- **Solution:** Define coherent information: $I(A \rangle B)_\rho = S(B)_\rho - S(AB)_\rho = -S(A | B)_\rho$
- **Interpretation:**
 - When $S(A | B) < 0$, $I(A \rangle B) > 0$
 - Coherent information quantifies how much quantum information is preserved through a noisy channel



Quantum Mutual Information

- **Definition:** For a bipartite quantum state ρ_{AB} , the mutual information is:

$$\begin{aligned} I(A : B)_\rho &= S(A)_\rho + S(B)_\rho - S(AB)_\rho \\ &= S(A)_\rho - S(A | B)_\rho \\ &= S(B)_\rho - S(B | A)_\rho \end{aligned}$$

- **Interpretation:** Measures total correlations (classical + quantum) between A and B .
- **Non-Negativity of Mutual Information:** $I(A : B) \geq 0$
- **Proof:** Conditioning does not increase entropy:
 $S(A | B) \leq S(A) \Rightarrow S(A) - S(A | B) \geq 0 \Rightarrow I(A : B) \geq 0$
- **Example:** Consider the Bell state: $|\Psi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$
 - **Properties:** Pure state: $S(AB) = 0$
 - **Marginals:** $\rho_A = \rho_B = \frac{1}{2}I \Rightarrow S(A) = S(B) = 1$
 - So, Mutual Information: $I(A : B) = 1 + 1 - 0 = 2$

Overview



- Motivation
- Quantum Entropy
- **Quantum Data Compression**
- Quantum Channels
- Quantum Capacity

Bit vs Qubit in Information Theory



- In Quantum IT, the **logarithm of the dimensionality** of the space is the measure of the information content of a system and plays the role of the logarithm of the size of a codebook in Classical IT
- CIT: a codebook $\mathcal{C} = \mathcal{A}^L$ with $|\mathcal{A}| = d$ contains d^L codewords and $\log |\mathcal{C}| = L \log d$, so that each codeword is represented by $L \log d$ bits.
- In QIT: we consider L -long sequences of states of the form $|\Psi\rangle = |\psi_{i_1}\rangle \otimes \dots \otimes |\psi_{i_L}\rangle$ where $|\psi_{i_i}\rangle \in \mathcal{H}$, so that $|\Psi\rangle \in \mathcal{H}^{\otimes L}$. If $\dim \mathcal{H} = d$, then $\log \dim \mathcal{H}^{\otimes L} = L \log d$.
- If we call *qudit* the basic Hilbert space $\mathcal{H}$, then the sequence $|\Psi\rangle$ is represented by L qudits, or equivalently by 1 qudit/symbol. For qubits: $\mathcal{H} = \mathcal{Q}$, with $\dim \mathcal{Q} = 2$

Source Coding Theorems



- **Question:** What is the minimum number of bit per symbol or qubit per symbol needed to represent a codeword or a state sequence? Answer: Source Coding Theorems
- **Shannon's Source Coding Theorem:** For classical i.i.d symbols with probability distribution $\{p_i\}$, as $L \rightarrow \infty$, a minimum of $H(A)$ bits/symbol are required for encoding
- **Schumacher's Source coding theorem:** In quantum case the probability distribution is encoded in density operator as $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ and determines the quantum entropy of a state $S(\rho)$, which becomes $LS(\rho)$ for the L -long sequence. In the limit the minimum number is given by a quantum entropy $S(\rho)$, expressed in qubit per symbol.

Quantum Data Compression



- **Aim:** Find an encoding protocol that maps the states of a Hilbert space into a new Hilbert space of reduced dimensionality, with the possibility of recovering the original states.
- Let $d = \dim \mathcal{H}$ and its L -extension $\mathcal{H}^{\otimes L}$ of dimension d^L where the source state sequences (quantum sourcewords) are defined.
- Map these sequences into quantum codewords in a new Hilbert space $\mathcal{H}_c$ s.t.
 $d_c := \dim \mathcal{H}_c < \dim \mathcal{H}^{\otimes L} = d^L$

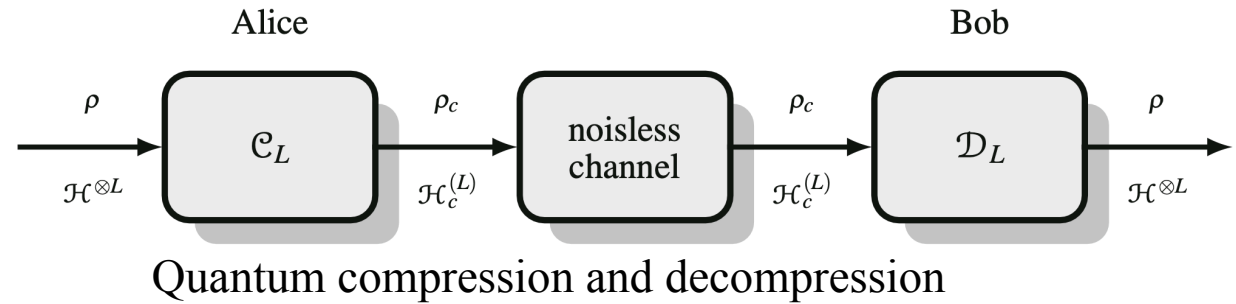


Schumacher's Compression Protocol

- Quantum compression and decompression:
 $\mathcal{C}_L : \mathcal{H}^{\otimes L} \rightarrow \mathcal{H}_c^{(L)}, \quad \mathcal{D}_L : \mathcal{H}_c^{(L)} \rightarrow \mathcal{H}^{\otimes L}$

- Compression Rate:

$$R := \frac{\log \dim \mathcal{H}_c^{(L)}}{\log \dim \mathcal{H}^{\otimes L}} = \frac{\log \dim \mathcal{H}_c^{(L)}}{L \log d}$$



- If $\mathcal{H}^{\otimes L}$ consists of L qubits ($d = 2$), then $R = \log \dim \mathcal{H}_c^{(L)} / L$, and $\dim \mathcal{H}_c^{(L)} = 2^{LR}$, thus R reads in qubits/symbol.
- $\mathcal{C}_L$ compresses ρ of $L \log d$ qubits into state ρ_c of LR qubits and $\mathcal{D}_L$ decompresses ρ_c into ρ .

- Fidelity:** $F(\rho, \sigma) = \left(\text{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$

s.t. $0 \leq F \leq 1$ and becomes $F = 1$ only when the two states coincide.

- Theorem:** (Schumacher's noiseless quantum channel coding) *Consider an IID quantum source with entropy $S(\rho)$ and rate R . Then, if $R > S(\rho)$ there exists an asymptotically reliable compression protocol (where $F \rightarrow 1$ as $L \rightarrow \infty$). Conversely, if $R < S(\rho)$ any compression protocol is not reliable.*